ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 8.01: Surface Integrals, Divergence Theorem & Stokes Theorem

Q1. Find nablacdotmathbf F at left 2, 1, 0right for mathbf F=langle y e^{x}, x z, y z^{2} rangle .

[x] A: e^{2}
[ ] B: 1 + e^{2}
[ ] C: - e^{2}
[ ] D: 2 e^{2}

Solution: nablacdotmathbf F=2 y z + y e^{x} ; at left 2, 1, 0right this is e^{2} .

Q2. Find nablacdotmathbf F at left 2, -2, 1right for mathbf F=langle x z, y z, x y rangle .

[ ] A: -4
[x] B: 2
[ ] C: 3
[ ] D: -2

Solution: nablacdotmathbf F=2 z ; at left 2, -2, 1right this is 2 .

Q3. Find nablacdotmathbf F at left 2, -2, 0right for mathbf F=langle y e^{x}, x z, y z^{2} rangle .

[ ] A: 1 - 2 e^{2}
[x] B: - 2 e^{2}
[ ] C: 2 e^{2}
[ ] D: - 4 e^{2}

Solution: nablacdotmathbf F=2 y z + y e^{x} ; at left 2, -2, 0right this is - 2 e^{2} .

Q4. Find nablacdotmathbf F at left 2, 1, -2right for mathbf F=langle x z, y z, x y rangle .

[ ] A: -3
[ ] B: 4
[x] C: -4
[ ] D: -8

Solution: nablacdotmathbf F=2 z ; at left 2, 1, -2right this is -4 .

Q5. Find nablatimesmathbf F at left 2, 0, 1right for mathbf F=langle y, - x, z^{2} rangle .

[ ] A: langle 0, 0, 2 rangle
[ ] B: langle 0, -2, 0 rangle
[x] C: langle 0, 0, -2 rangle
[ ] D: langle 1, 1, -1 rangle

Solution: nablatimesmathbf F=langle 0, 0, -2 rangle ; at left 2, 0, 1right this is langle 0, 0, -2 rangle .

Q6. Find nablacdotmathbf F at left -2, 1, -2right for mathbf F=langle x^{2}, y^{2}, z^{2} rangle .

[ ] A: 9
[ ] B: -5
[ ] C: 6
[x] D: -6

Solution: nablacdotmathbf F=2 x + 2 y + 2 z ; at left -2, 1, -2right this is -6 .

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Q7. Find nablatimesmathbf F at left -2, -2, 2right for mathbf F=langle y e^{x}, x z, y z^{2} rangle .

[ ] A: langle -6, 0, -2 + e^{-2} rangle

[x] B: langle 6, 0, 2 - e^{-2} rangle

[ ] C: langle 0, 2 - e^{-2}, 6 rangle

[ ] D: langle 7, 1, 3 - e^{-2} rangle

Solution: nablatimesmathbf F=langle - x + z^{2}, 0, z - e^{x} rangle ; at left -2, -2, 2right this is langle 6, 0, 2 - e^{-2} rangle .

Q8. Find nablacdotmathbf F at left 0, -2, 0right for mathbf F=langle y, - x, z^{2} rangle .

[ ] A: -2

[x] B: 0

[ ] C: 1

[ ] D: 2

Solution: nablacdotmathbf F=2 z ; at left 0, -2, 0right this is 0 .